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Stahl SM , Eisenach JC , Taylor CP , et al. The diverse therapeutic actions of pregabalin: is a single mechanism responsible for several pharmacologic activities . *Trends Pharmacological Sci* 2013 ;**34** (6):332–9 .

Propranolol

Therapeutics

Brands

- Inderal
- Inderal LA
- InnoPran XL
- *see index for additional brand names*

Generic?

Yes

Class

- Beta blocker, antihypertensive

Commonly Prescribed for

(bold for FDA approved)

- **Migraine prophylaxis**
- **Essential tremor**
- **Hypertension**
- **Angina pectoris due to coronary atherosclerosis**

- **Cardiac arrhythmias (including supraventricular arrhythmias, ventricular tachycardia, digitalis intoxication)**
- **Myocardial infarction**
- **Hypertrophic subaortic stenosis**
- **Pheochromocytoma**
- Akathisia (antipsychotic induced)
- Parkinsonian tremor
- Violence, aggression
- PTSD, prophylactic
- Generalized anxiety disorder (GAD)
- Prevention of variceal bleeding
- Congestive heart failure
- Tetralogy of Fallot
- Hyperthyroidism (adjunctive)

How the Drug Works

- For migraine, proposed mechanisms include inhibition of the adrenergic pathway, interaction with the serotonin system and receptors, inhibition of nitric oxide synthesis, and normalization of contingent negative variation
- For tremor, antagonism of peripheral beta 2 receptors is the proposed mechanism

- For PTSD, blockade of beta 1 adrenergic receptors may theoretically prevent fear conditioning and reconsolidation of fear
- For violence/aggression, the mechanism is poorly established; presumed to be related to central actions at beta adrenergic and serotonin receptors

How Long Until It Works

- For migraine, can begin to work within 2 weeks, but may take up to 3 months on a stable dose to see full effect
- For tremor, can begin to work within days

If It Works

- For migraine, the goal is a 50% or greater decrease in migraine frequency or severity; consider tapering or stopping if headaches remit for more than 6 months
- For tremor, can cause reduction in the severity of tremor, allowing greater functioning with daily activities and clearer speech
- For PTSD, may theoretically block the effects of stress from prior traumatic experiences
- For aggression, may reduce aggression, agitation, or uncooperativeness

If It Doesn't Work

- Increase to highest tolerated dose

- For migraine, address other issues such as medication overuse or other coexisting medical disorders; consider changing to another drug or adding a second drug
- For tremor, coadministration with primidone up to 250 mg/day can augment response; second-line medications include benzodiazepines, gabapentin, topiramate, methazolamide, nadolol, and botulinum toxin (useful for voice and hand tremor); alternative treatments include caffeine and hand weights
- For patients with truly refractory tremor, thalamotomy or deep brain stimulation of the ventral intermediate nucleus of the thalamus is an option
- For PTSD, consider initiating first-line pharmacotherapy (SSRI, SNRI) and psychotherapy
- For violence/aggression, switch to another agent, e.g., valproate or an antipsychotic



Best Augmenting Combos for Partial Response or Treatment Resistance

- Migraine: for some patients, low-dose polytherapy with 2 or more drugs may be better tolerated and more effective than high-dose monotherapy; may use propranolol in combination with anticonvulsants, antidepressants, natural products, and non-pharmacologic treatments, such as biofeedback, to improve headache control
- For tremor, can combine with primidone or second-line medications

- For aggression and violence, can combine with valproate and/or antipsychotics

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- By blocking beta adrenergic receptors, it can cause dizziness, bradycardia, and hypotension

Notable Side Effects

- Bradycardia, hypotension, hyper- or hypoglycemia, weight gain
- Bronchospasm, cold/flu symptoms, sinusitis, pneumonias
- Dizziness, vertigo, fatigue/tiredness, depression, sleep disturbances
- Sexual dysfunction, decreased libido, dysuria, urinary retention, joint pain
- Exacerbation of symptoms in peripheral vascular disease and Raynaud's syndrome



Life-Threatening or Dangerous Side Effects

- In acute congestive heart failure, may further depress myocardial contractility

- Can blunt premonitory symptoms of hypoglycemia in diabetes and mask clinical signs of hyperthyroidism
- Nonselective beta blockers such as propranolol can inhibit bronchodilation, making them contraindicated in asthma, severe COPD
- Do not use in pheochromocytoma unless alpha blockers are already being used
- Risk of excessive myocardial depression in general anesthesia

Weight Gain



- Many experience and/or can be significant in amount

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Lower dose, change to an extended-release formulation, or switch to another agent

Best Augmenting Agents for Side Effects

- When patients have significant benefit from beta blocker therapy but hypotension limits treatment, consider alpha agonists (midodrine) or

volume expanders (fludrocortisones) for symptomatic relief

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 40–400 mg/day

Dosage Forms

- Tablet 10 mg, 20 mg, 40 mg, 60 mg, 80 mg, 90 mg
- Extended-release capsule 60 mg, 80 mg, 120 mg, 160 mg
- Oral solution 4 mg/mL, 8 mg/mL
- Injection 1 mg/mL

How to Dose

- Migraine: initial dose 40 mg/day in divided doses or once daily in extended-release preparations; gradually increase over days to weeks until effective dose is reached; maximum dose 400 mg/day
- Tremor: initial dose 40 mg twice per day; dosage may be gradually increased as needed to 120–320 mg/day in 2 to 3 divided doses
- PTSD: effective dose varies greatly; up to 240 mg/day has been used

- Aggression: same as migraine; up to 400 mg/day if tolerated and effective



Dosing Tips

- For extended-release capsules, give once daily at bedtime consistently, with or without food
- Doses above 120 mg had no additional antihypertensive effect in clinical trials
- High doses may be effective in some patients with tremor, migraine, or aggression/violence

Overdose

- Bradycardia, hypotension, low-output heart failure, shock, seizures, coma, hypoglycemia, apnea, cyanosis, respiratory depression, and bronchospasm

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Should not be discontinued abruptly; instead gradually reduce dosage over 1–2 weeks
- May exacerbate angina, and there are reports of tachyarrhythmias or myocardial infarction with rapid discontinuation in patients with cardiac disease

Pharmacokinetics

- Half-life 3–5 hours (immediate-release) or 8–11 hours (extended-release)



Drug Interactions

- Cimetidine, oral contraceptives, ciprofloxacin, hydralazine, hydroxychloroquine, loop diuretics, certain SSRIs (with CYP 2D6 metabolism), and phenothiazines can increase levels and/or effects of propranolol
- Use with calcium channel blockers can be synergistic or additive, use with caution
- Barbiturates, penicillins, rifampin, calcium and aluminum salts, thyroid hormones, and cholestyramine can decrease effects of beta blockers
- NSAIDs, sulfinpyrazone, and salicyclates inhibit prostaglandin synthesis and may inhibit the antihypertensive activity of beta blockers

- Propranolol can increase adverse effects of gabapentin and benzodiazapines
- Propranolol can increase levels of lidocaine, resulting in toxicity, and increase the anticoagulant effect of warfarin
- Increased postural hypotension with prazosin and peripheral ischemia with ergot alkaloids
- Sudden discontinuation of clonidine while on beta blockers or when stopping together can cause life-threatening increases in blood pressure

Other Warnings/Precautions

- May elevate blood urea, serum transaminases, alkaline phosphatase, and lactate dehydrogenase (LDH)
- Rare development of antinuclear antibodies (ANAs)
- May worsen symptoms of myasthenia gravis
- Can lower intraocular pressure, interfering with glaucoma screening test

Do Not Use

- If patient has bradycardia, greater than first-degree heart block, or cardiogenic shock
- If patient has bronchial asthma or severe COPD
- If there is a proven allergy to propranolol

Special Populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- Use with caution with severe impairment; dose reduction may be necessary

Cardiac Impairment

- Do not use in acute shock, myocardial infarction, hypotension, and greater than first-degree heart block, but indicated in clinically stable patients post-myocardial infarction to reduce risk of re-infarction starting 1–4 weeks after event

Elderly

- Use with caution
- May increase risk of stroke



Children and Adolescents

- Usual dose in children is 2–4 mg/kg in 2 divided doses; maximum 16 mg/kg/day
- Clinical trials for migraine prophylaxis did not include children



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- May reduce perfusion of the placenta
- Use only if potential benefits outweigh the potential risks to the fetus

Breast Feeding

- Some drug is found in mother's breast milk
 - Due to high lipid solubility, propranolol is found in breast milk more than many other beta blockers
- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child

The Art of Psychopharmacology

Potential Advantages

- Patients who do not respond to or tolerate other options
- Patients with autonomic hyperactivity

Potential Disadvantages

- Multiple potential undesirable adverse effects, including bradycardia, hypotension, and fatigue

Primary Target Symptoms

- Migraine frequency and severity
- Tremor
- Effects of stress from prior traumatic experience
- Aggression, agitation

Pearls

- Often used in combination with other drugs in migraine, which may allow patients to better tolerate medications that cause tremor, such as valproate
- May worsen depression, but helpful for anxiety
- 50–70% of patients with essential tremor receive some relief, usually with about 50% improvement or greater
- Propranolol may theoretically block the effects of stress from prior traumatic experiences, but this is unproven and data thus far are mixed

Suggested Reading

Brunet A , Poundja J , Tremblay J , et al. Trauma reactivation under the influence of propranolol decreases posttraumatic stress symptoms and disorders: 3 open-label trials . *J Clin Psychopharmacol* 2011 ;**31** (4):547–50 .

Cohen H , Kaplan Z , Koresh O , et al. Early post-stressor intervention with propranolol is ineffective in preventing posttraumatic stress responses in an animal model for PTSD . *Eur Neuropsychopharmacol* 2011 ;**21** (3):230–40 .

Fleminger S , Greenwood RJ , Oliver DL . Pharmacological management for agitation and aggression in people with acquired brain injury . *Cochrane Database Syst Rev* 2003 ;(1):CD003299 .

Lyons KE , Pahwa R . Pharmacotherapy of essential tremor: an overview of existing and upcoming agents . *CNS Drugs* 2008 ;**22** (12):1037–45 .

Silberstein SD . Preventive migraine treatment . *Neurol Clin* 2009 ;**27** (2):429–43 .

Protriptyline

Therapeutics

Brands

- Triptil
- Vivactil
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine reuptake inhibitor
- Tricyclic antidepressant (TCA)
- Predominantly a norepinephrine/noradrenaline reuptake inhibitor

Commonly Prescribed for

(bold for FDA approved)

- **Mental depression**
- Treatment-resistant depression

How the Drug Works

- Boosts neurotransmitter norepinephrine/noradrenaline
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, protriptyline can increase dopamine neurotransmission in this part of the brain
- A more potent inhibitor of norepinephrine reuptake pump than serotonin reuptake pump (serotonin transporter)
- At high doses may also boost neurotransmitter serotonin and presumably increase serotonergic neurotransmission

How Long Until It Works

- * Some evidence it may have an early onset of action with improvement in activity and energy as early as 1 week
- Onset of therapeutic actions usually not immediate, but often delayed 2 to 4 weeks
- If it is not working within 6 to 8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- Lithium, buspirone, thyroid hormone

Tests

- Baseline ECG is recommended for patients over age 50
- * Since tricyclic and tetracyclic antidepressants are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management
- * Monitor weight and BMI during treatment
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes,

or dyslipidemia, or consider switching to a different antidepressant

- ECGs may be useful for selected patients (e.g., those with personal or family history of QTc prolongation; cardiac arrhythmia; recent myocardial infarction; uncompensated heart failure; or taking agents that prolong QTc interval such as pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin, etc.)
- Patients at risk for electrolyte disturbances (e.g., patients on diuretic therapy) should have baseline and periodic serum potassium and magnesium measurements

Side Effects

How Drug Causes Side Effects

- * Anticholinergic activity for protriptyline may be more potent than for some other TCAs and may explain sedative effects, dry mouth, constipation, blurred vision, tachycardia, and hypotension
- Sedative effects and weight gain may be due to antihistamine properties
- Blockade of alpha 1 adrenergic receptors may explain dizziness, sedation, and hypotension
- Cardiac arrhythmias, especially in overdose, may be caused by blockade of ion channels

Notable Side Effects

- Blurred vision, constipation, urinary retention, increased appetite, dry mouth, nausea, diarrhea, heartburn, unusual taste in mouth, weight gain
- Fatigue, weakness, dizziness, sedation, headache, anxiety, nervousness, restlessness
- Sexual dysfunction (impotence, change in libido)
- Sweating, rash, itching

Life-Threatening or Dangerous Side Effects

- Paralytic ileus, hyperthermia (TCAs + anticholinergic agents)
- Lowered seizure threshold and rare seizures
- Orthostatic hypotension, sudden death, arrhythmias, tachycardia
- QTc prolongation
- Hepatic failure, drug-induced parkinsonism
- Increased intraocular pressure
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Many experience and/or can be significant in amount

- Can increase appetite and carbohydrate craving

Sedation



- Many experience and/or can be significant in amount
- * Not as sedating as other TCAs; more likely to be activating than other TCAs

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Switch to an SSRI or newer antidepressant

Best Augmenting Agents for Side Effects

- Trazodone or a hypnotic for insomnia
- Benzodiazepines for agitation and anxiety
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 15–40 mg/day in 3–4 divided doses

Dosage Forms

- Tablet 5 mg, 10 mg

How to Dose

- Initial 15 mg/day in divided doses; increase morning dose as needed; maximum dose 60 mg/day



Dosing Tips

* Be aware that among this class of agents (tricyclic/tetracyclic antidepressants), protriptyline has uniquely low dosing (15–40 mg/day for protriptyline compared to 75–300 mg/day for most other tricyclic/tetracyclic antidepressants)

* Be aware that among this class of agents (tricyclic/tetracyclic antidepressants), protriptyline has uniquely frequent dosing (3–4 times a day compared to once daily for most other tricyclic/tetracyclic antidepressants)

- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder, and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- Death may occur; CNS depression, convulsions, cardiac dysrhythmias, severe hypotension, ECG changes, coma

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects
- Even with gradual dose reduction some withdrawal symptoms may appear within the first 2 weeks
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation
- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

Pharmacokinetics

- Substrate for CYP450 2D6
- Half-life approximately 74 hours

Drug Interactions

- Tramadol increases the risk of seizures in patients taking TCAs
- Use of TCAs with anticholinergic drugs may result in paralytic ileus or hyperthermia
- Fluoxetine, paroxetine, bupropion, duloxetine, and other 2D6 inhibitors may increase TCA concentrations
- Cimetidine may increase plasma concentrations of TCAs and cause anti-cholinergic symptoms
- Phenothiazines or haloperidol may raise TCA blood concentrations
- May alter effects of antihypertensive drugs; may inhibit hypotensive effects of clonidine
- Use with sympathomimetic agents may increase sympathetic activity
- Methylphenidate may inhibit metabolism of TCAs
- Activation and agitation, especially following switching or adding antidepressants, may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of protriptyline

Other Warnings/Precautions

- Add or initiate other antidepressants with caution for up to 2 weeks after discontinuing protriptyline